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SAMPLE DELIVERABLE · ILLUSTRATIVE EXAMPLE

AI Workflow Audit: Goods receipt and deviation handling

How I would audit one regulated workflow, decide where AI belongs and where it does not, and turn the result into a pilot a QA team can sign off.

This is an example, not a client project. The company, volumes and findings are fictional and simplified. The workflow is typical for GDP-regulated pharmaceutical logistics, an environment I have rolled out systems in. It shows the structure and depth of a real audit deliverable.

Workflow

Inbound goods receipt, deviation recording and QA decision

Setting

Fictional mid-sized pharmaceutical logistics provider, GDP and ISO 9001

Audit format

5 business days: interviews, document review, process walk-through

01 · EXECUTIVE SUMMARY

One pilot, two AI steps, zero AI decisions

Goods receipt is slowed down by manual re-typing of document data and by deviations that live in free text, email and a spreadsheet. AI can take over the reading and structuring work. Release decisions stay with QA, exactly as GDP requires.

Key findings

#	Finding	Why it matters
1	Batch, expiry and quantity are typed by hand from delivery notes and certificates of analysis into the WMS.	Transcription errors reach stock records and traceability.
2	Deviations are recorded as free text in a shared spreadsheet, with photos sent by email.	Incomplete records, no trend analysis, slow audit preparation.
3	QA learns about deviations by email and has no view of open cases or their age.	Goods sit in quarantine longer than needed.
4	Temperature excursions are checked by eye in the logger portal.	A rules problem, not an AI problem. Fix with thresholds.

Recommendation

PILOT

Structured goods receipt check

AI extracts document data into a fixed schema. Deterministic rules compare it with the purchase order and advance shipping notice. Mismatches open a pre-filled deviation record. Warehouse confirms, QA decides.

EXPLICITLY OUT OF SCOPE

No automated release

AI never releases, rejects or quarantines goods and never closes a deviation. Every AI output is editable, logged and confirmed by a named person.

Decision needed

Approve a 2 week baseline measurement and a test set of historical documents, followed by a 2 to 4 week build sprint and a shadow-mode run before any go-live decision.

02 · SCOPE AND CONTEXT

What was looked at

Assumed setting

- One GDP-certified warehouse, ambient (15 to 25 °C) and cold chain (2 to 8 °C) goods
- Around 40 inbound deliveries per day from roughly 120 suppliers
- Warehouse team on two shifts, QA team of three
- Mixed document quality: PDFs, scans, paper delivery notes

Systems in use

- Warehouse management system (WMS), the system of record
- Shared mailbox for supplier documents and advance shipping notices
- Spreadsheet deviation log, paper inspection checklist
- Temperature logger portal from the carrier

All figures are assumptions for this example.

How the audit was run

Day	Activity	Output
1	Kick-off with operations and QA, collect documents, SOPs, sample records	Scope, stakeholder map
2	Walk-through of 3 live deliveries, interviews with warehouse lead and QA	Current process map
3	Review of 50 historical receipts and 20 deviation records	Bottlenecks, data quality notes
4	Opportunity scoring, risk assessment with QA	Opportunity matrix, pilot design
5	Read-out and decision meeting	This report, roadmap

03 · CURRENT PROCESS

Nine steps, four systems, one spreadsheet

01	**Pre-advice** Supplier emails an advance shipping notice and documents. Documents arrive in a shared mailbox, not linked to the order.	Admin
02	**Arrival and document check** Delivery note, certificate of analysis and temperature data are checked against the purchase order. Manual comparison on paper.	Warehouse
03	**Physical inspection** Packaging, labels, quantity, batch and expiry. Paper checklist, photos on a phone.	Warehouse
04	**Temperature review** Logger data is checked for excursions. Read by eye in a separate portal.	Warehouse
05	**Booking in quarantine** Batch, expiry and quantity are typed into the WMS. Main source of transcription errors.	Warehouse
06	**Deviation recording** Any mismatch is written into the spreadsheet log, photos emailed to QA. Free text, missing fields, no categories.	Warehouse
07	**QA assessment** QA reviews evidence and decides: release, reject, return or investigate. No overview of open cases or their age.	QA
08	**Supplier complaint and CAPA** Where needed, a complaint letter is drafted and corrective actions tracked. Letters written from scratch each time.	QA
09	**Release to saleable stock** Status changed in the WMS after QA sign-off.	QA

04 · OPPORTUNITY MATRIX

Where AI helps, and where it should stay out

Each candidate is scored on value, feasibility with the available data, and regulatory risk. Some problems are better solved with plain rules. Saying so is part of the audit.

Candidate	Value	Feasibility	Risk	Verdict
A. Document extraction Delivery note and certificate of analysis into structured fields, compared with the order.	High	High	Medium	Pilot
B. Deviation intake Free text and photos into a structured deviation record with a suggested category.	High	Medium	Medium	Pilot
C. Supplier complaint draft First draft of the complaint letter from the deviation record.	Medium	High	Low	Phase 2
D. Temperature excursion check Flag readings outside the product's storage range.	Medium	High	Low	Rules, not AI
E. Automated release decision Let the system release or reject goods.	n/a	n/a	High	Excluded

Why E is excluded: under GDP the release decision belongs to accountable people in QA. AI can prepare the evidence. It must not make or imply the decision.

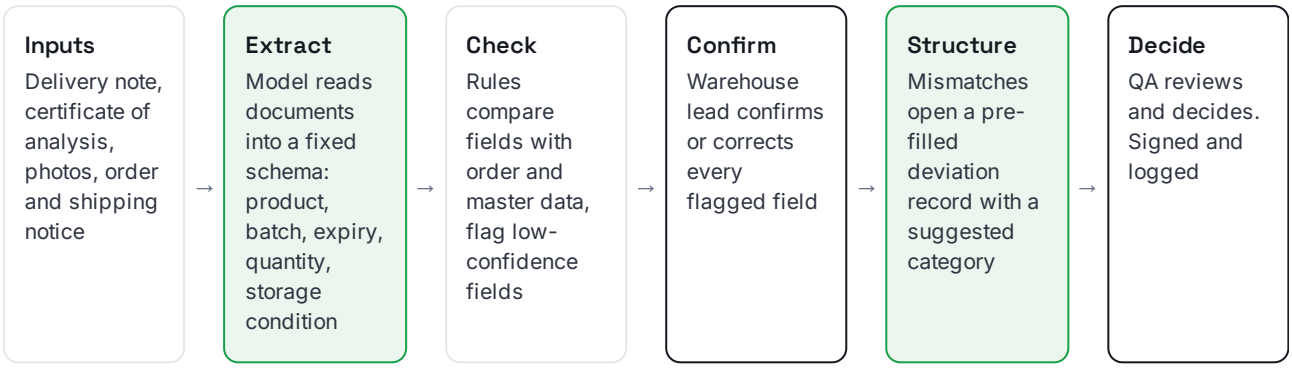
Why D is rules: storage ranges are fixed per product. A deterministic threshold check is cheaper, explainable and easier to validate than a model.

05 · PILOT DESIGN

Structured goods receipt check

The pilot combines candidates A and B. Each step has one owner: the model, a rule or a person.

AI step · Deterministic rule or system · Human decision



The WMS stays the system of record. For a lightweight version, an intake app with role-gated status tracking works too; my public FlowLogics demo shows that pattern.

What AI does

- Reads documents and fills a fixed schema
- Reports a confidence level per field
- Turns free text and photos into a structured deviation draft
- Suggests a deviation category

What stays deterministic

- Comparison with order, shipping notice and master data
- Storage range and expiry checks
- Status changes in the WMS

Controls

- No automated release, rejection or deviation closure
- Every AI output editable; edits and confirmations logged with user and time
- Low-confidence fields always shown to a person
- Versioned prompts and schema, change control with QA
- Model hosting and data processing agreed up front, client data not used for model training
- Role-based access, least privilege
- Validation approach proportionate to risk, agreed with QA before go-live

06 · MEASUREMENT

Baseline first, targets second

No improvement figures are promised in an audit. Targets are set after a two week baseline, with the client, on these measures.

Measure	How it is measured
Time from arrival to booked in quarantine	Timestamps from WMS and receipt log
Time from deviation found to QA decision	Deviation record timestamps
Transcription errors in batch, expiry, quantity	Sample of records checked against source documents
Complete deviation records (all required fields)	Share of records passing a completeness check
Warehouse and QA time per receipt	Short time study on a sample of deliveries

How the model is tested before anyone relies on it

- A test set of around 100 anonymised historical documents, labelled by the client team
- Field-level accuracy per document type, with a pass threshold agreed with QA before the build
- Shadow mode: the system runs next to the current process, people work as usual, results are compared
- Go-live only if the agreed threshold is met and QA signs off

07 · ROADMAP AND RISKS

From audit to a decision in about seven weeks

WEEK 0	WEEKS 1 TO 2	WEEKS 2 TO 4	WEEKS 5 TO 6	WEEK 7
Audit	**Baseline**	**Build sprint**	**Shadow mode**	**Decision**
This report and pilot scope	Measures and labelled test set	Extraction, rules, review screens, deviation draft	Run in parallel, compare results	Go-live, adjust or stop, with QA

Risks and mitigations

Risk	Mitigation
Poor scan quality lowers extraction accuracy	Low-confidence fields always go to a person; scanning guideline for suppliers
Team over-trusts AI suggestions	Confirmation required per flagged field, spot checks by QA in shadow mode
Supplier document formats change	Test set extended with each new supplier format, monitored error rate
Data protection and hosting concerns	Hosting and data processing agreed before the build, no training on client data
Pilot adds work instead of removing it	Shadow mode measures time per receipt; stop criteria agreed up front

What a real audit needs from the client

- Three 45 minute interviews: warehouse lead, QA, operations manager
- Relevant SOPs, 50 recent receipts and 20 deviation records, anonymised where needed
- Read access to the deviation log and a WMS walkthrough

> Want this for one of your workflows? Book a 20 minute intro call at **mazuryk.dev/services**. Prepared by Roman Mazuryk, Technical Project Manager, 12+ years of system implementation in MedTech, pharma and logistics.